

Mathematical Representation

Let

$$X_i$$

be a minority-class sample.

Choose one of its nearest neighbors

$$X_{nn}$$

Generate a synthetic sample as

$$X_{\text{new}} = X_i + \lambda(X_{nn} - X_i)$$

where

- $0 \leq \lambda \leq 1$ is a random number.

This places the new sample at a random position on the line segment connecting the two minority samples.

Numerical Example

Original Dataset

Sample	Feature 1	Feature 2	Class
A	2	3	Minority
B	3	4	Minority
C	8	9	Majority
D	9	10	Majority
E	10	11	Majority

Suppose SMOTE chooses samples **A** and **B**.

Take

$$\lambda = 0.5$$

Then

$$\begin{aligned} X_{\text{new}} &= (2, 3) + 0.5[(3, 4) - (2, 3)] \\ &= (2, 3) + 0.5(1, 1) \\ &= (2.5, 3.5) \end{aligned}$$

The new synthetic sample is

Feature 1	Feature 2	Class
2.5	3.5	Minority

The minority class now has an additional sample without duplicating existing data.

Advantages of SMOTE

- Balances imbalanced datasets effectively.
- Reduces model bias toward the majority class.
- Generates synthetic samples instead of duplicating data.
- Improves recall and F1-score for minority classes.
- Works well with many machine learning algorithms.
- Helps reduce overfitting caused by simple oversampling.

Limitations of SMOTE

- May generate noisy or overlapping synthetic samples.
- Can increase training time due to a larger dataset.
- Less effective when the minority class contains many outliers.
- Not suitable for every dataset without careful parameter tuning.
- May not preserve complex class boundaries.

Applications

- Fraud detection
- Credit risk assessment
- Medical diagnosis
- Disease prediction
- Spam email detection
- Network intrusion detection
- Customer churn prediction
- Fault detection in manufacturing
- Insurance claim analysis